

Partial Markov categories with Exact Observations

Elena Di Lavore

February 12, 2026

This note defines partial Markov categories with exact observations and proves some facts about them. They seem to be a minimal setting where Bayes Theorem holds. Exact observations turn out to be property rather than structure.

Exact observations. Intuitively, an exact observation of a state a in a copy-discard category sets the value of a wire to a . In principle, there could be exact observations of nondeterministic states as well, but it is not clear how they should behave.

Definition 0.1. A state $a: I \rightarrow A$ in a copy-discard category $\mathbb{C}$ has an *exact observation* if there is a predicate $a^\circ: A \rightarrow I$ satisfying the two equations below.

$$\begin{aligned} A \text{ --- } \bullet \text{ --- } A \text{ (with a loop labeled } a^\circ \text{)} &= A \text{ --- } [a^\circ] \text{ --- } [a] \text{ --- } A \\ [a] \text{ --- } [a^\circ] &= [a] \text{ --- } \bullet \end{aligned}$$

A copy-discard category $\mathbb{C}$ has *exact observations* if all deterministic states have an exact observation.

A synthetic version of Bayes' theorem is immediate from the axioms.

Theorem 0.2 (Synthetic Bayes' Theorem). *In a partial Markov category with exact observations, an exact observation $a^\circ: A \rightarrow I$ from a prior distribution $\sigma: I \rightarrow X$ through a channel $f: X \rightarrow A$ determines an update that is proportional to the evaluation of the channel f in the state a , $f_\sigma^\dagger(a)$.*

$$\sigma \text{ --- } \bullet \text{ --- } X \text{ (with a loop labeled } f \text{ and } a^\circ \text{)} = \sigma \text{ --- } [f] \text{ --- } [a^\circ] \text{ --- } [a] \text{ --- } [f_\sigma^\dagger] \text{ --- } X$$

Proof. The proof uses (i) bayesian inversions and (ii) the first axiom of exact observations.

$$\begin{array}{c}
\begin{array}{c} \sigma \end{array} \begin{array}{c} \bullet \end{array} \begin{array}{c} \xrightarrow{\quad} X \\ \xrightarrow{\quad} f \end{array} \begin{array}{c} a^\circ \end{array} \\
\stackrel{(i)}{=} \begin{array}{c} \sigma \end{array} \begin{array}{c} \xrightarrow{\quad} f \end{array} \begin{array}{c} \bullet \end{array} \begin{array}{c} \xrightarrow{\quad} f_\sigma^\dagger \\ \xrightarrow{\quad} a^\circ \end{array} \begin{array}{c} X \end{array} \\
\stackrel{(ii)}{=} \begin{array}{c} \sigma \end{array} \begin{array}{c} \xrightarrow{\quad} f \end{array} \begin{array}{c} a^\circ \end{array} \quad \begin{array}{c} a \end{array} \begin{array}{c} \xrightarrow{\quad} f_\sigma^\dagger \end{array} \begin{array}{c} X \end{array}
\end{array}$$

□

Exact observations are a property of copy-discard categories: when they exist they are unique.

Lemma 0.3. *Exact observations are unique: a state in a copy-discard category $\mathbb{C}$ has at most one exact observation.*

Proof. Consider a state $a: I \rightarrow A$ and suppose that there are two predicates, a° and $a^\bullet$, that satisfy the axioms of exact observations.

$$\begin{aligned}
a^\bullet &= \nu \circ (\varepsilon \otimes a^\bullet) = a^\bullet \circ a \circ \varepsilon = a^\bullet \circ a \circ a^\circ \\
&= \nu \circ (a^\circ \otimes a^\bullet) = \nu \circ (a^\bullet \otimes a^\circ) = a^\circ \circ a \circ a^\bullet \\
&= a^\circ \circ a \circ \varepsilon = \nu \circ (\varepsilon \otimes a^\circ) = a^\circ
\end{aligned}$$

We used the axioms of exact observations, and commutativity and unitality of the comonoid structure. □

The next result justifies the name “exact observations”: they are always deterministic. Interestingly, this does not depend on the associated state being deterministic.

Lemma 0.4. *The exact observation of a state $a: I \rightarrow A$ in a copy-discard category $\mathbb{C}$ is deterministic, $a^\circ = \nu \circ (a^\circ \otimes a^\circ)$.*

Proof. We use the axioms of exact observations and unitality of the comonoid structure.

$$\begin{aligned}
\nu \circ (a^\circ \otimes a^\circ) &= a^\circ \circ a \circ a^\circ = a^\circ \circ a \circ \varepsilon \\
&= \nu \circ (\varepsilon \otimes a^\circ) = a^\circ
\end{aligned}$$

□

Copy-discard functors preserve exact observations.

Lemma 0.5. *Let $F: \mathbb{C} \rightarrow \mathbb{D}$ be a strict copy-discard functor between two strict copy-discard categories, $\mathbb{C}$ and $\mathbb{D}$. If $a^\circ: A \rightarrow I$ is the exact observation of $a: I \rightarrow A$ in $\mathbb{C}$, then $F(a^\circ)$ is the exact observation of $F(a)$ in $\mathbb{D}$. In particular, if $\mathbb{D}$ has exact observations, then $F(a^\circ) = (Fa)^\circ$ for all states $a: I \rightarrow A$ in $\mathbb{C}$.*

Proof. We use that F is a strict copy-discard functor and that a° is an exact observation in $\mathbb{C}$. We check the first axiom of exact observations.

$$\begin{aligned}
& \nu_{FA} \circ (\mathbb{1}_{FA} \otimes F(a^\circ)) \\
&= F(\nu_A) \circ F(\mathbb{1}_A \otimes a^\circ) \\
&= F(\nu_A) \circ (\mathbb{1}_A \otimes a^\circ) \\
&= F(a^\circ \circ a) \\
&= F(a^\circ) \circ F(a)
\end{aligned}$$

We check the second axiom of exact observations.

$$\begin{aligned}
& F(a) \circ F(a^\circ) \\
&= F(a \circ a^\circ) \\
&= F(a \circ \varepsilon_A) \\
&= F(a) \circ F(\varepsilon_A) \\
&= F(a) \circ \varepsilon_{FA}
\end{aligned}$$

If $\mathbb{D}$ has exact observations, then $F(a^\circ) = (Fa)^\circ$ for all states $a: I \rightarrow A$ in $\mathbb{C}$ because exact observations are unique. $\square$

Future work. We plan construct the free partial Markov category with exact observations over a Markov category. We plan to investigate the relationship between exact observations of nondeterministic states and interventions.